

Railway Animal Collision Prevention Device

Team GuardianX



Department of Electronic & Telecommunication Engineering

University of Moratuwa

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Index No	Name
230033J	AMARASINGHE A.A.D.K
230208C	GUNASEKARA H.M
230508V	RAHUL B
230526A	RANAWAKA R.A.C.D

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1. Problem Description

1.1. Problem

Animals colliding with trains on the railway tracks have been a problem in Sri Lanka for a very long time without any kind of sustainable solution, especially elephants. When an animal collides with a train, it causes a huge loss in life and financial loss. This greatly disrupts the government of Sri Lanka by raising the need to fix the train and train tracks after an accident.

1.2. Expected Goal

Our goal is to provide a long-term sustainable solution to prevent these accidents for the Sri Lankan government, which consequently saves them a lot of money. Our solution works on all trains currently operating in Sri Lanka.

1.3. Solution

Our solution consists of a transmitter and a receiver. The receivers will be placed on all trains, and the transmitters will be deployed in previously known locations where accidents occur frequently. The receiver constantly observes the environment it is placed in with the help of PIR and microwave sensors, and broadcasts a signal to the receiver to let the train driver know about possible animals being present on the railway tracks. With the help of the received signal, the train driver can stop the train in time and wait until the receiver stops receiving the signal and continues on its way.

1.4. Justification of Selection

Since there are no current solutions available for this problem, our system is perfect to prevent accidents along railway tracks with animals. We can reduce the loss of lives and money using this system.

2. Product Feasibility

2.1. Technical Feasibility

We have successfully made our project technically viable and obtained all the necessary resources. We used a PIR sensor and a microwave sensor to detect the animals' presence around the railway tracks. We used the HC-12 transceiver module to transmit and receive signals on both sides of our system.

We used the ATmega328 micro controller to control our system. We have included 11.1V rechargeable batteries on both the receiver and the transmitter for long-time sustainable use. We designed our own PCB and enclosure for our system of devices and imported all the electronic components needed in our device from LCSC and printed our PCB from JLCPCB. Rysera innovations helped us to 3D print our enclosure for both the receiver and transmitter.

2.2. Hardware Feasibility

The hardware components used in our project:

- ATmega328p micro controllers
- HC-SR501 Pyroelectric Infrared PIR Motion Sensor Detector Modules
- HC-12 Transceiver Modules
- RCWL-0516 Microwave Radar Modules
- LM2596 Buck Converters
- 11.1V 18650 2400mAh Batteries

2.3. Software Feasibility

Software	Programming Language	Purpose
Arduino	C++	Program the ATmega328p Micro Controller
SolidWorks	—————	Design the enclosures for our system
Altium Designer	—————	Design the PCBs for our system

3. Product Applications

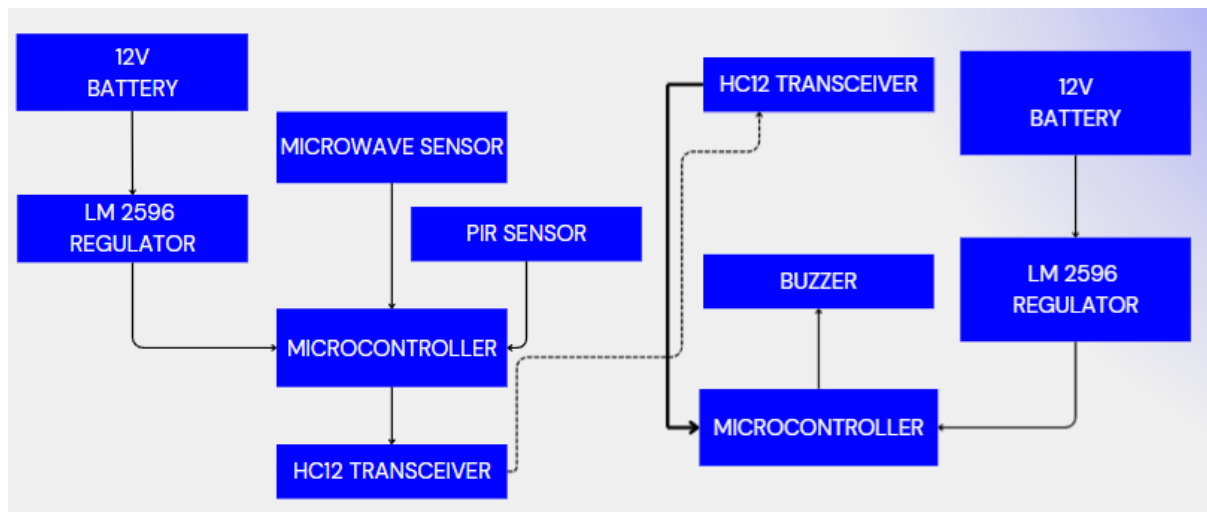
Main purpose of our system of devices is to accurately scan the environment near the railway tracks and detect if an animal is blocking the railway track or not. If an animal is indeed blocking the railway track then our system sends out a signal with a radius of 1.5 km to be picked up by our receiver placed on any one of the oncoming trains. Our receiver activates the buzzer to let the driver know about the situation on the upcoming railway track.

Both devices can be recharged using a dc jack. There is an LED placed on the receiver which will blink when the battery is low to let the driver know. Total of 2 LEDs are on the receiver, one lets the driver know that the receiver is active and the other lights up when the receiver is recharging.

4. Product Architecture

- Power Supply - Both devices are powered through 11.1V 18650 2400mAh rechargeable batteries, and supports recharging.
- ATmega328p Chip - This controls all the sensors and the transceiver modules and coordinates them to operate properly as a system.
- Voltage Regulator - LM2596 Step-Down DC-DC Buck Converter which regulates the input voltage and regulates it to constant 5V DC voltage.
- PIR sensor - detects changes in infrared (heat) radiation emitted by objects in its field of view.
- RCWL0516 microwave radar - utilizes the Doppler effect to detect movement.
- HC-12 Transceiver Module - Used on our devices to transmit data over long range to reach the trains.

Block Diagram



5. Product Enclosure

5.1. Initial Sketch

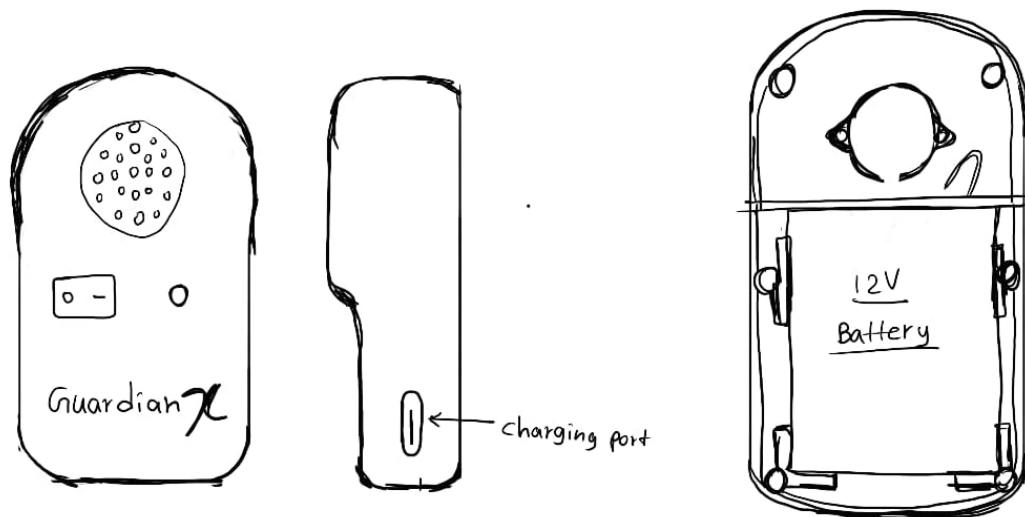


Figure 1: Receiver

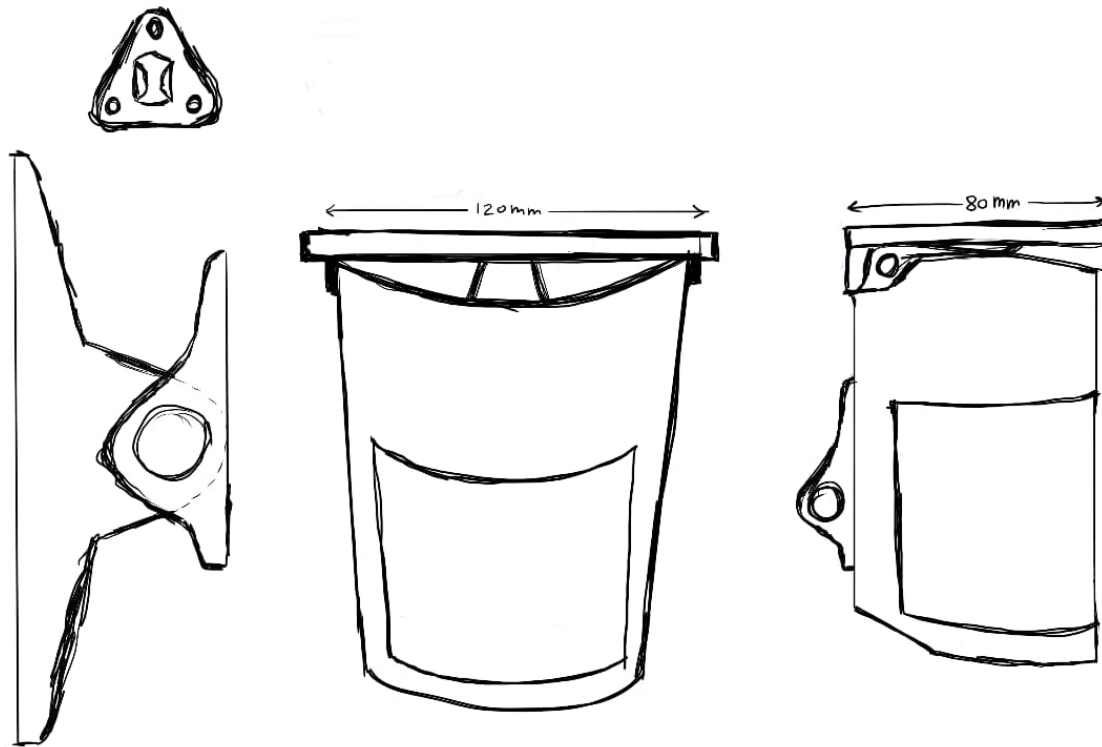
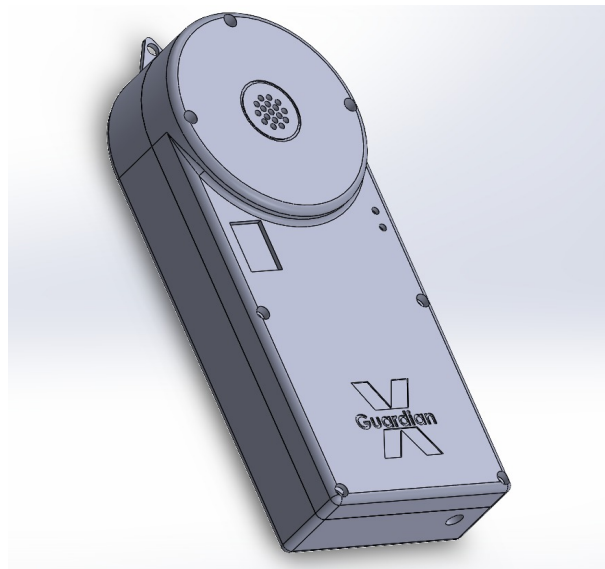


Figure 2: Transmitter

5.2. Receiver



5.3. Transmitter

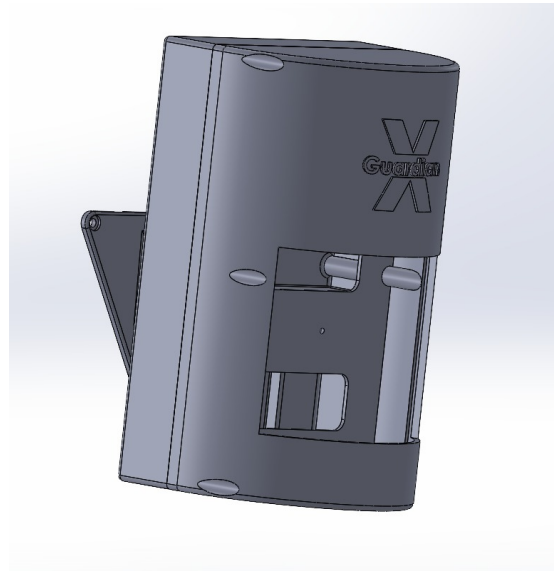


Figure 4: Receiver

6.2. PCB

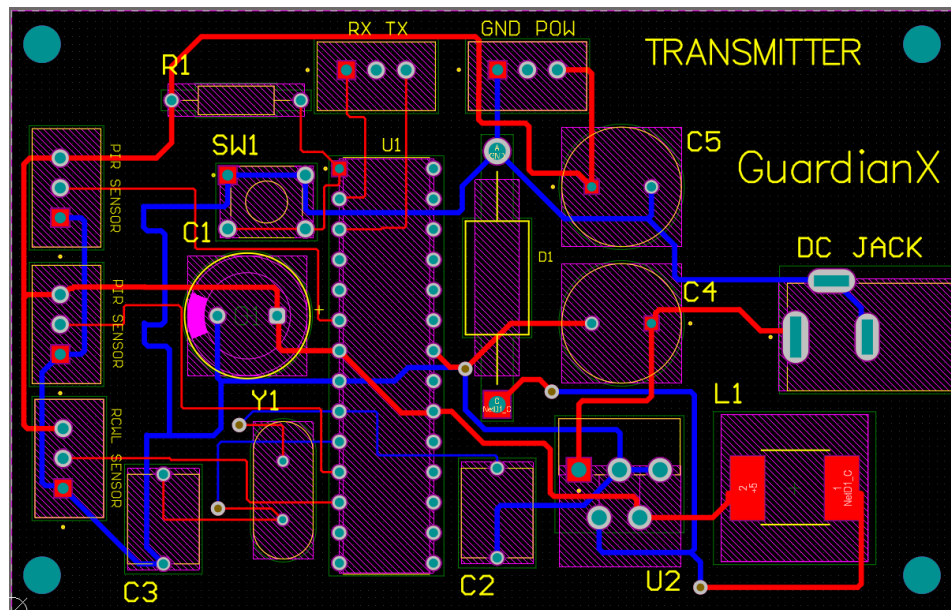


Figure 5: Transmitter

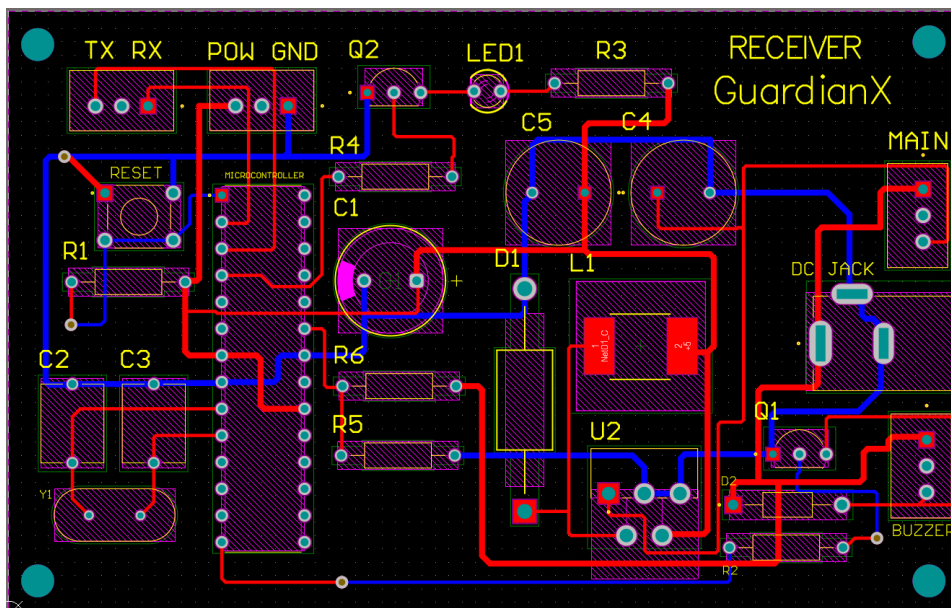


Figure 6: Receiver

7. Final Product



Figure 7: Overview

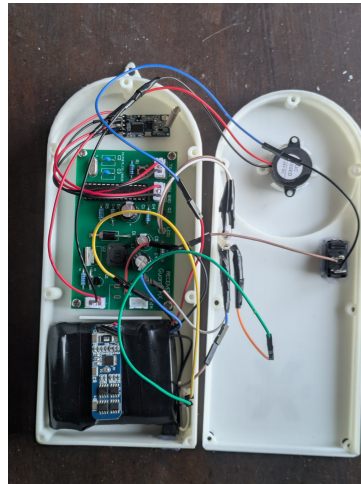


Figure 8: Receiver

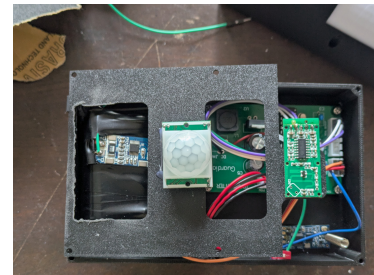


Figure 9: Transmitter

8. Market Analysis

Our product designed specifically to be deployed along railway tracks where our device would be hung on the towers placed along the railway for animal detection. Our product is a system developed for the government to utilize and prevent animal and railway collisions and reduce financial losses. It's cost-effective and proved to be reliable most of the time giving accurate results. With the help of the government, we can start mass production of our product and deploy everywhere in Sri Lanka.

9. Task Allocation

Name	Task
AMARASINGHE A.A.D.K	Enclosure Design, Soldering
GUNASEKARA H.M	Circuit Analysis, Assembling
RAHUL B	PCB Design, Circuit Design
RANAWAKA R.A.C.D	Micro Controller Programming, Soldering

10. Final Budget

Item	Price (LKR)
Components	6,800.00
PCB	3,300.00
Tax	5,500.00
Modules	4,720.00
Enclosure	13,115.00
Soldering Equipment	1,675.00
Screws & Bolts	500.00
Total	35,610.00

A. Appendices

PIR Sensor Data Sheet <https://components101.com/sensors/hc-sr501-pir-sensor>

RCWL-0516 Data Sheet <https://www.handsontec.com/dataspecs/sensor/radar-sensor.pdf>

ATMega328p Data Sheet <https://ww1.microchip.com/downloads/en/DeviceDoc/Atmel-7810-Automotive-Microcontrollers-ATmega328P>

Buzzer Data Sheet <https://docs.rs-online.com/1d9b/A700000010343551.pdf>

HC-12 Transceiver Data Sheet <https://www.alldatasheet.com/datasheet-pdf/pdf/1242982/ETC1/12.html>

B. Acknowledgements

We extend our heartfelt thanks to the dedicated staff of the Analog Lab for their outstanding support, particularly in accommodating our after-hours lab sessions, which played a crucial role in the progress of our project. We also wish to acknowledge the administrative team at ENTC for their prompt and effective assistance in providing essential equipment and resources. Their efforts were instrumental in the successful execution of our experiments. Our sincere gratitude goes to Dr. Pasqual and our fellow batchmates, whose guidance, encouragement, and collaboration have been deeply appreciated throughout this journey. Thank you all for your continued support and contribution.